SHRED-MPC loop

MODEL PREDICTIVE CONTROLLER

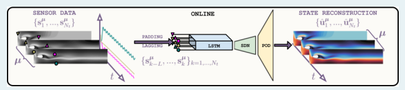
INPUT SENSORS

- Sparse pressure
  $\overline{p_s}(t-L+1 : t)$
- Rotational velocities
  $\overline{\omega}(t-L+1 : t)$

L = lag length (timesteps)

SHRED SURROGATE

Control-conditioned predictor



Pressure field ⟶ lift

$$e_k = \hat{y}_k - r_k$$

ŷ: mean predicted pressure lift

r: reference lift

Pressure proxy, not total lift

OPTIMIZER

Minimize total cost

$$J_k = J_{\text{track}} + J_{\text{osc}} + J_{\Delta\omega}$$

Tracking · oscillation · control change

Select the lowest-cost profile

Candidate rotational velocity

PHYSICAL PLANT

OpenFOAM CFD

Advance the flow and return new sparse pressure data

At each MPC update: predict many candidate futures with SHRED, apply one control segment in CFD, then replan.

SHRED-ROM online panel (cropped): Matteo Tomasetto – Stefano Riva (2024), MIT. Upstream architecture illustration.